Linear Approximation and L'Hôpital's Rule

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

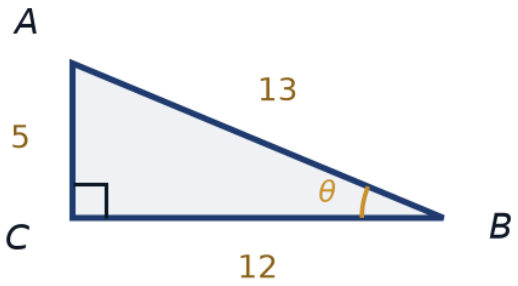
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

2 [Foundational] • Target: L'Hôpital's Rule for 0/0 Forms

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.



3 [Foundational] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{10})$.

4 [Foundational] • Target: Comparing Relative Growth Rates of Functions

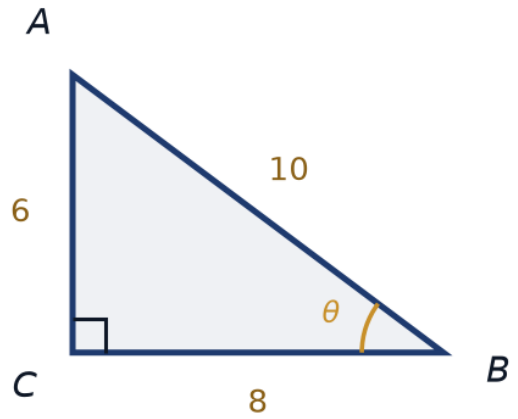
Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{5})$.

5 [Foundational] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

6 [Foundational] • Target: L'Hôpital's Rule for 0/0 Forms

Use linearization (the tangent line) of $f(x) = 3\sqrt{x}$ at $x = 5$ to approximate $f(5 + \frac{1}{5})$.



© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 5$ to approximate $f(5 + \frac{1}{5})$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

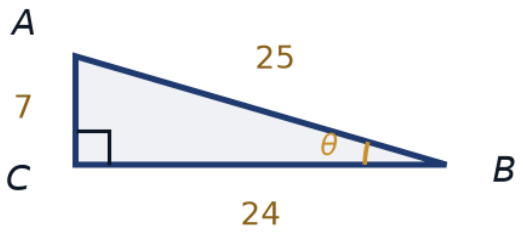
Use linearization (the tangent line) of $f(x) = 5\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{10})$.

9 [Intermediate] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 3$ to approximate $f(3 - \frac{1}{5})$.

10 [Intermediate] • Target: L'Hôpital's Rule for 0/0 Forms

Use linearization (the tangent line) of $f(x) = 5\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.



11 [Intermediate] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Use linearization (the tangent line) of $f(x) = 4\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{5})$.

12 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

13 [Intermediate] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 3$ to approximate $f(3 - \frac{1}{10})$.

14 [Intermediate] • Target: L'Hôpital's Rule for 0/0 Forms

Use linearization (the tangent line) of $f(x) = 5\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{5})$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

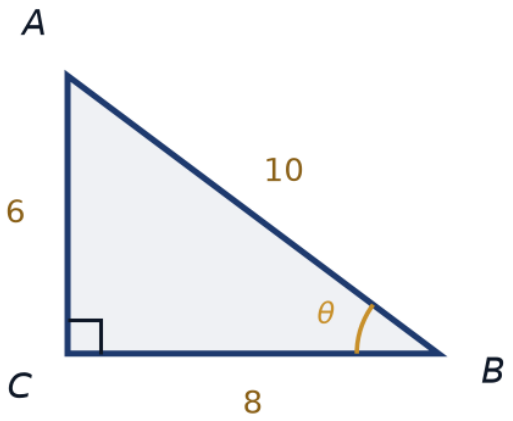
15 [Challenging] • Target: L'Hôpital's Rule for Infinity/Infinity Forms • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Comparing Relative Growth Rates of Functions • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{10})$.



17 [Challenging] • Target: Tangent Line Approximation (Linearization) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Use linearization (the tangent line) of $f(x) = 5\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{10})$.

18 [Challenging] • Target: L'Hôpital's Rule for 0/0 Forms • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Use linearization (the tangent line) of $f(x) = 3\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{5})$.

19 [Challenging] • Target: L'Hôpital's Rule for Infinity/Infinity Forms • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 5$ to approximate $f(5 - \frac{1}{5})$.

20 [Challenging] • Target: Comparing Relative Growth Rates of Functions • Cross-Lesson: 8.01 Area Between Curves

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{5})$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Tangent Line Approximation (Linearization)

What is the local linear approximation $L(x)$ of $f(x) = \sqrt{x}$ at $x = 25$?

a. $L(x) = 5 + \frac{1}{10}(x - 25)$

b. $L(x) = 5 + \frac{1}{5}(x - 25)$

c. $L(x) = 5 + \frac{1}{2}(x - 25)$

d. $L(x) = \frac{1}{10}x$

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Tangent Line Approximation (Linearization)

Using the linearization of $f(x) = \sqrt{x}$ at $a = 25$, approximate $\sqrt{26}$.

a. 5.10

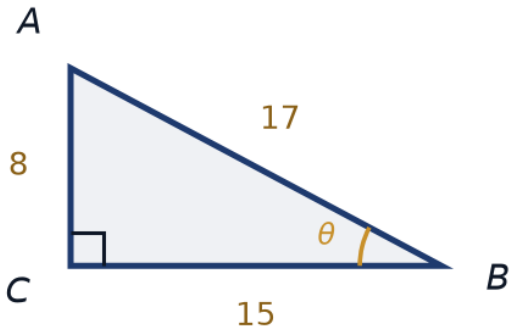
b. 5.05

c. 5.20

d. 5.01

23 [Foundational] • Target: Tangent Line Approximation (Linearization)

Is the tangent line approximation for $f(x) = \sqrt{x}$ an overestimate or underestimate, and why?



a. An overestimate because $f''(x) < 0$ (concave down)

b. An underestimate because $f''(x) > 0$ (concave up)

c. Exact because f is monotonic

d. Depends on the step size

24 [Foundational] • Target: L'Hôpital's Rule for 0/0 Forms

Evaluate $\lim_{x \to 0} \frac{e^{2x} - 1}{\sin(3x)}$.

a. $\frac{2}{3}$

b. $\frac{3}{2}$

c. 0

d. ∞

25 [Foundational] • Target: L'Hôpital's Rule for 0/0 Forms

Evaluate $\lim_{x \to 1} \frac{\ln x}{x^2 - 1}$.

a. $\frac{1}{2}$

b. 1

c. 2

d. 0

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: L'Hôpital's Rule for 0/0 Forms

Evaluate $\lim_{x \to 0} \frac{1 - \cos(4x)}{x^2}$.

a. 8

b. 4

c. 2

d. 16

27 [Foundational] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Evaluate $\lim_{x \to \infty} \frac{\ln(x)}{x}$.

a. 0

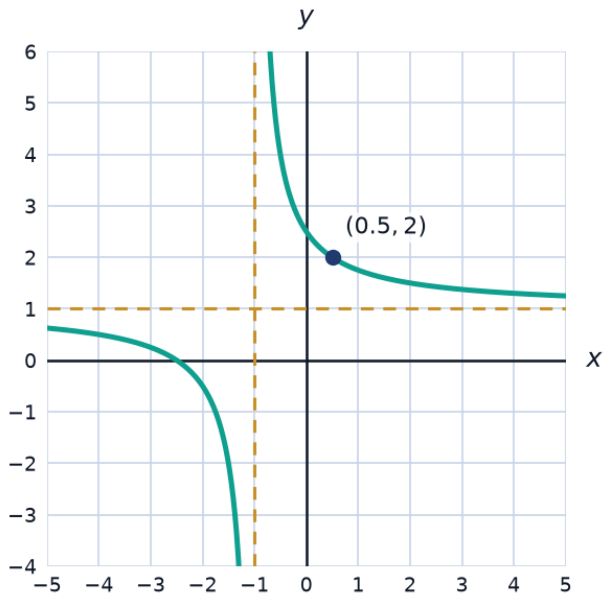
b. 1

c. ∞

d. Undefined

28 [Foundational] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Evaluate $\lim_{x \to \infty} \frac{x^2}{e^x}$.



a. 0

b. 1

c. 2

d. ∞

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Which requirement MUST be verified before applying L'Hôpital's Rule to $\lim_{x \to c} \frac{f(x)}{g(x)}$?

a. The limit produces an indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$

b. $f(c) = 0$ only

c. $f'(c) = 0$

d. $g(x) > 0$ for all x

30 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

As $x \to \infty$, which of the following functions grows the fastest?

a. e^x

b. x^{10}

c. $x \ln x$

d. 2^x

31 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

Evaluate $\lim_{x \to \infty} \frac{(\ln x)^5}{\sqrt{x}}$.

a. 0

b. ∞

c. 1

d. 5

32 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

If $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$, what does this conclude about their growth rates?

a. $f(x)$ grows faster than $g(x)$

b. $g(x)$ grows faster than $f(x)$

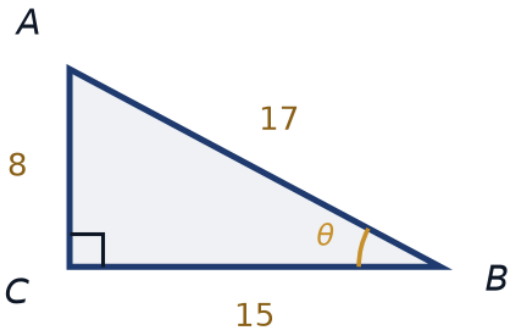
c. They grow at the same rate

d. Neither function grows

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.



a. $-1 + \frac{39\sqrt{2}}{40}$

b. $\sqrt{2}$

c. $1 + \frac{39\sqrt{2}}{40}$

d. $\frac{39\sqrt{2}}{40}$

34 [Intermediate] • Target: L'Hôpital's Rule for 0/0 Forms

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{10})$.

a. $\sqrt{3}$

b. $\frac{61\sqrt{3}}{60}$

c. $1 + \frac{61\sqrt{3}}{60}$

d. $-1 + \frac{61\sqrt{3}}{60}$

35 [Intermediate] • Target: L'Hôpital's Rule for Infinity/Infinity Forms

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

a. $\sqrt{2}$

b. $-1 + \frac{41\sqrt{2}}{40}$

c. $\frac{41\sqrt{2}}{40}$

d. $1 + \frac{41\sqrt{2}}{40}$

36 [Intermediate] • Target: Comparing Relative Growth Rates of Functions

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{5})$.

a. $1 + \frac{21\sqrt{2}}{20}$

b. $\frac{21\sqrt{2}}{20}$

c. $\sqrt{2}$

d. $-1 + \frac{21\sqrt{2}}{20}$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Tangent Line Approximation (Linearization)

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 3$ to approximate $f(3 + \frac{1}{10})$.

a. $-1 + \frac{61\sqrt{3}}{10}$

b. $\frac{61\sqrt{3}}{10}$

c. $6\sqrt{3}$

d. $1 + \frac{61\sqrt{3}}{10}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: L'Hôpital's Rule for 0/0 Forms • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Use linearization (the tangent line) of $f(x) = 4\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{10})$.

a. $\frac{41\sqrt{2}}{10}$

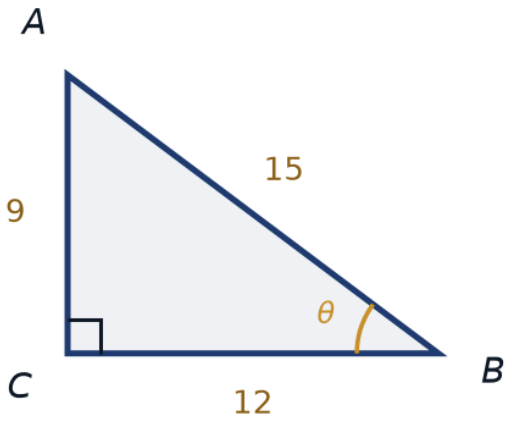
b. $1 + \frac{41\sqrt{2}}{10}$

c. $4\sqrt{2}$

d. $-1 + \frac{41\sqrt{2}}{10}$

39 [Challenging] • Target: L'Hôpital's Rule for Infinity/Infinity Forms • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Use linearization (the tangent line) of $f(x) = 4\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.



a. $\frac{39\sqrt{2}}{10}$

b. $-1 + \frac{39\sqrt{2}}{10}$

c. $1 + \frac{39\sqrt{2}}{10}$

d. $4\sqrt{2}$

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Comparing Relative Growth Rates of Functions • Cross-Lesson: 8.01 Area Between Curves

Use linearization (the tangent line) of $f(x) = 4\sqrt{x}$ at $x = 2$ to approximate $f(2 + \frac{1}{5})$.

a. $1 + \frac{21\sqrt{2}}{5}$

b. $-1 + \frac{21\sqrt{2}}{5}$

c. $4\sqrt{2}$

d. $\frac{21\sqrt{2}}{5}$

41 [Challenging] • Target: Tangent Line Approximation (Linearization) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 4$ to approximate $f(4 + \frac{1}{10})$.

a. $\frac{263}{20}$

b. $\frac{223}{20}$

c. $\frac{243}{20}$

d. 12

42 [Challenging] • Target: L'Hôpital's Rule for 0/0 Forms • Cross-Lesson: 1.01 Limits Graphically and Numerically

Use linearization (the tangent line) of $f(x) = 6\sqrt{x}$ at $x = 4$ to approximate $f(4 - \frac{1}{5})$.

a. $\frac{107}{10}$

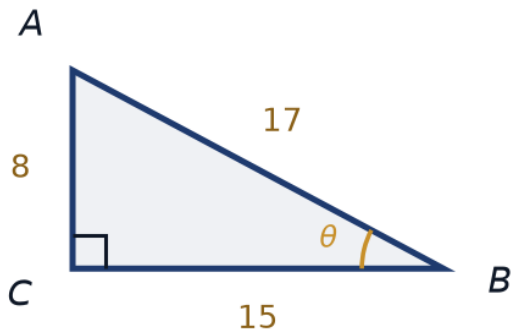
b. 12

c. $\frac{117}{10}$

d. $\frac{127}{10}$

43 [Challenging] • Target: L'Hôpital's Rule for Infinity/Infinity Forms • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Use linearization (the tangent line) of $f(x) = 7\sqrt{x}$ at $x = 1$ to approximate $f(1 - \frac{1}{10})$.



a. $\frac{153}{20}$

b. $\frac{133}{20}$

c. 7

d. $\frac{113}{20}$

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Comparing Relative Growth Rates of Functions • Cross-Lesson: 2.01 The Definition of the Derivative

Use linearization (the tangent line) of $f(x) = 2\sqrt{x}$ at $x = 4$ to approximate $f(4 - \frac{1}{10})$.

a. $\frac{59}{20}$

b. $\frac{99}{20}$

c. $\frac{79}{20}$

d. 4

45 [Challenging] • Target: Tangent Line Approximation (Linearization) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 4$ to approximate $f(4 - \frac{1}{10})$.

a. $\frac{119}{40}$

b. $\frac{39}{40}$

c. 2

d. $\frac{79}{40}$

© ClipSAT — clipsat.org | All Rights Reserved